

1. Akamu's Table (20 marks)

Adapted from "Compilation of Russian Olympiad problems between 1965 and 1975" by Vladimir Belikov, Elena Muravenko, Alekseev Mikhail Egorovic

"Did you know that the Hawaiian language and the Maori language spoken by the Maori people of your country are related?" said Akamu, the tourist guide, to Tia, the tourist from New Zealand, who was visiting Hawaii for the first time. "That's impossible!" exclaimed Tia. "The two archipelagos are separated by 7500 km of water. How could two languages spoken in islands so far apart could be similar?" Akamu smiled. "Well, it is not only these two languages. There are at least 30 other languages spread across the islands of the Pacific Ocean that evolved from the same common ancestor. Linguists call this the *Polynesian family of languages*". He took out a piece of paper and wrote down the words for the cardinal numbers (first, second, third etc.) in 5 Polynesian languages that he could recall. He handed it over to Tia and said "You don't have to believe me; see it for yourself. How would a set of unrelated languages have so similar words for the cardinal numbers?"

Tia looked at Akamu's table with utmost surprise (' and wh are specific consonants):

<u>Language</u>	1	2	3	4	5	6	7	8	9
Hawaiian	<i>kahi</i>	<i>lua</i>	(a)	<i>ha</i>	<i>lima</i>	<i>ono</i>	<i>hiku</i>	<i>walu</i>	(b)
Maori	<i>tahi</i>	<i>rua</i>	<i>toru</i>	<i>wha</i>	(c)	<i>ono</i>	<i>whitu</i>	<i>waru</i>	<i>iwa</i>
Nuku Hiva	<i>tahi</i>	(d)	<i>to'u</i>	<i>ha</i>	(e)	<i>ono</i>	(f)	<i>va'u</i>	(g)
Rarotonga	<i>ta'i</i>	(h)	(i)	<i>'a</i>	<i>rima</i>	<i>ono</i>	<i>'itu</i>	<i>varu</i>	<i>iva</i>
Samoa	<i>tasi</i>	<i>lua</i>	(j)	(k)	<i>lima</i>	<i>ono</i>	<i>fitu</i>	(l)	<i>iva</i>

It was not hard to see the pattern. Akamu had forgotten some of the words – the gray cells in the table. Tia could fill in the gaps simply by studying the pattern. She showed Akamu her guesses, and Akamu exclaimed "Woah! You got everything right. When did you learn all these languages?" "It's actually simple", said Tia.

Assignment 1 [Answer Booklet]: Guess the 12 missing words.

Assignment 2: Explain the rules that you used to convert words from one language to another. Here is one example rule (as well as a hint) for you: 'the consonant *l* in Hawaiian is transformed to *r* in Maori'.
